

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
from scipy import stats
```

```
In [2]: df=pd.read_csv("D:\StudentPerformance.csv")
```

```
In [3]: df
```

```
Out[3]:
```

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.0	64	84.0	2020	2
1	female	71.0	80.0	76	86.0	2018	3
2	female	64.0	81.0	66	81.0	2020	2
3	male	71.0	85.0	77	96.0	2018	1
4	male	68.0	86.0	76	NaN	2021	3
5	female	94.0	86.0	61	100.0	2019	1
6	male	75.0	79.0	66	-99.0	2020	1
7	female	NaN	NaN	66	95.0	2019	3
8	male	66.0	88.0	66	88.0	2020	3
9	male	70.0	79.0	61	87.0	2021	2
10	female	-99.0	80.0	65	85.0	2021	1
11	male	76.0	84.0	-99	NaN	2020	2
12	female	74.0	79.0	79	98.0	2019	2

In [4]: `df.isnull()`

Out[4]:

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	True	False	False
5	False	False	False	False	False	False	False
6	False	False	False	False	False	False	False
7	False	True	True	False	False	False	False
8	False	False	False	False	False	False	False
9	False	False	False	False	False	False	False
10	False	False	False	False	False	False	False
11	False	False	False	False	True	False	False
12	False	False	False	False	False	False	False

In [5]: `series = pd.isnull(df["reading score"])`
`df[series]`

Out[5]:

	gender	math score	reading score	writing score	placement score	club join year	placement offer
7	female	NaN	NaN	66	95.0	2019	3

In [6]: `df.notnull()`

Out[6]:

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	True	True	True	True	True	True	True
1	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True
3	True	True	True	True	True	True	True
4	True	True	True	True	False	True	True
5	True	True	True	True	True	True	True
6	True	True	True	True	True	True	True
7	True	False	False	True	True	True	True
8	True	True	True	True	True	True	True
9	True	True	True	True	True	True	True
10	True	True	True	True	True	True	True
11	True	True	True	True	False	True	True
12	True	True	True	True	True	True	True

```
In [7]: seseries = pd.notnull(df["reading score"])
df[series]
```

```
Out[7]:
```

	gender	math score	reading score	writing score	placement score	club join year	placement offer
7	female	NaN	NaN	66	95.0	2019	3

```
In [8]: from sklearn.preprocessing import LabelEncoder
le = LabelEncoder ()
df['gender'] = le.fit_transform(df['gender'])
newdf=df
df
```

```
Out[8]:
```

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	0	63.0	84.0	64	84.0	2020	2
1	0	71.0	80.0	76	86.0	2018	3
2	0	64.0	81.0	66	81.0	2020	2
3	1	71.0	85.0	77	96.0	2018	1
4	1	68.0	86.0	76	NaN	2021	3
5	0	94.0	86.0	61	100.0	2019	1
6	1	75.0	79.0	66	-99.0	2020	1
7	0	NaN	NaN	66	95.0	2019	3
8	1	66.0	88.0	66	88.0	2020	3
9	1	70.0	79.0	61	87.0	2021	2
10	0	-99.0	80.0	65	85.0	2021	1
11	1	76.0	84.0	-99	NaN	2020	2
12	0	74.0	79.0	79	98.0	2019	2

```
In [9]: df['gender'].replace({1: "male", 0: "female"}, inplace=True)
df
```

```
Out[9]:
```

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.0	64	84.0	2020	2
1	female	71.0	80.0	76	86.0	2018	3
2	female	64.0	81.0	66	81.0	2020	2
3	male	71.0	85.0	77	96.0	2018	1
4	male	68.0	86.0	76	NaN	2021	3
5	female	94.0	86.0	61	100.0	2019	1
6	male	75.0	79.0	66	-99.0	2020	1
7	female	NaN	NaN	66	95.0	2019	3
8	male	66.0	88.0	66	88.0	2020	3
9	male	70.0	79.0	61	87.0	2021	2
10	female	-99.0	80.0	65	85.0	2021	1
11	male	76.0	84.0	-99	NaN	2020	2
12	female	74.0	79.0	79	98.0	2019	2

```
In [10]: ndf=df
ndf.fillna(0)
```

```
Out[10]:
```

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.0	64	84.0	2020	2
1	female	71.0	80.0	76	86.0	2018	3
2	female	64.0	81.0	66	81.0	2020	2
3	male	71.0	85.0	77	96.0	2018	1
4	male	68.0	86.0	76	0.0	2021	3
5	female	94.0	86.0	61	100.0	2019	1
6	male	75.0	79.0	66	-99.0	2020	1
7	female	0.0	0.0	66	95.0	2019	3
8	male	66.0	88.0	66	88.0	2020	3
9	male	70.0	79.0	61	87.0	2021	2
10	female	-99.0	80.0	65	85.0	2021	1
11	male	76.0	84.0	-99	0.0	2020	2
12	female	74.0	79.0	79	98.0	2019	2

```
In [11]: m_v=df['reading score'].mean()
df['reading score'].fillna(value=m_v, inplace=True)
df
```

```
Out[11]:
```

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.000000	64	84.0	2020	2
1	female	71.0	80.000000	76	86.0	2018	3
2	female	64.0	81.000000	66	81.0	2020	2
3	male	71.0	85.000000	77	96.0	2018	1
4	male	68.0	86.000000	76	NaN	2021	3
5	female	94.0	86.000000	61	100.0	2019	1
6	male	75.0	79.000000	66	-99.0	2020	1
7	female	NaN	82.583333	66	95.0	2019	3
8	male	66.0	88.000000	66	88.0	2020	3
9	male	70.0	79.000000	61	87.0	2021	2
10	female	-99.0	80.000000	65	85.0	2021	1
11	male	76.0	84.000000	-99	NaN	2020	2
12	female	74.0	79.000000	79	98.0	2019	2

```
In [12]: ndf.replace(to_replace = np.nan, value = -99)
```

```
Out[12]:
```

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.000000	64	84.0	2020	2
1	female	71.0	80.000000	76	86.0	2018	3
2	female	64.0	81.000000	66	81.0	2020	2
3	male	71.0	85.000000	77	96.0	2018	1
4	male	68.0	86.000000	76	-99.0	2021	3
5	female	94.0	86.000000	61	100.0	2019	1
6	male	75.0	79.000000	66	-99.0	2020	1
7	female	-99.0	82.583333	66	95.0	2019	3
8	male	66.0	88.000000	66	88.0	2020	3
9	male	70.0	79.000000	61	87.0	2021	2
10	female	-99.0	80.000000	65	85.0	2021	1
11	male	76.0	84.000000	-99	-99.0	2020	2
12	female	74.0	79.000000	79	98.0	2019	2

In [13]: `ndf.dropna()`

Out[13]:

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.0	64	84.0	2020	2
1	female	71.0	80.0	76	86.0	2018	3
2	female	64.0	81.0	66	81.0	2020	2
3	male	71.0	85.0	77	96.0	2018	1
5	female	94.0	86.0	61	100.0	2019	1
6	male	75.0	79.0	66	-99.0	2020	1
8	male	66.0	88.0	66	88.0	2020	3
9	male	70.0	79.0	61	87.0	2021	2
10	female	-99.0	80.0	65	85.0	2021	1
12	female	74.0	79.0	79	98.0	2019	2

In [14]: `ndf.dropna(how = 'all')`

Out[14]:

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.000000	64	84.0	2020	2
1	female	71.0	80.000000	76	86.0	2018	3
2	female	64.0	81.000000	66	81.0	2020	2
3	male	71.0	85.000000	77	96.0	2018	1
4	male	68.0	86.000000	76	NaN	2021	3
5	female	94.0	86.000000	61	100.0	2019	1
6	male	75.0	79.000000	66	-99.0	2020	1
7	female	NaN	82.583333	66	95.0	2019	3
8	male	66.0	88.000000	66	88.0	2020	3
9	male	70.0	79.000000	61	87.0	2021	2
10	female	-99.0	80.000000	65	85.0	2021	1
11	male	76.0	84.000000	-99	NaN	2020	2
12	female	74.0	79.000000	79	98.0	2019	2

```
In [15]: ndf.dropna(axis = 1)
```

```
Out[15]:
```

	gender	reading score	writing score	club join year	placement offer
0	female	84.000000	64	2020	2
1	female	80.000000	76	2018	3
2	female	81.000000	66	2020	2
3	male	85.000000	77	2018	1
4	male	86.000000	76	2021	3
5	female	86.000000	61	2019	1
6	male	79.000000	66	2020	1
7	female	82.583333	66	2019	3
8	male	88.000000	66	2020	3
9	male	79.000000	61	2021	2
10	female	80.000000	65	2021	1
11	male	84.000000	-99	2020	2
12	female	79.000000	79	2019	2

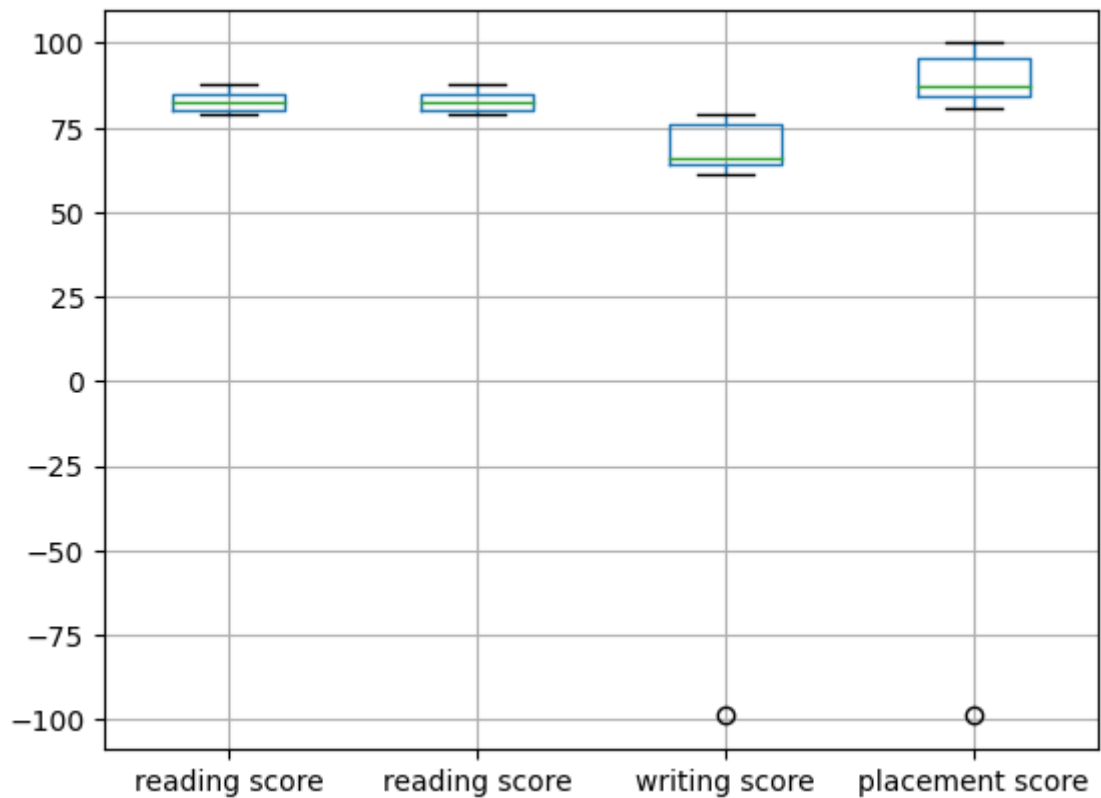
```
In [16]: new_data =ndf.dropna (axis = 0, how = 'any' )  
new_data
```

```
Out[16]:
```

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.0	64	84.0	2020	2
1	female	71.0	80.0	76	86.0	2018	3
2	female	64.0	81.0	66	81.0	2020	2
3	male	71.0	85.0	77	96.0	2018	1
5	female	94.0	86.0	61	100.0	2019	1
6	male	75.0	79.0	66	-99.0	2020	1
8	male	66.0	88.0	66	88.0	2020	3
9	male	70.0	79.0	61	87.0	2021	2
10	female	-99.0	80.0	65	85.0	2021	1
12	female	74.0	79.0	79	98.0	2019	2

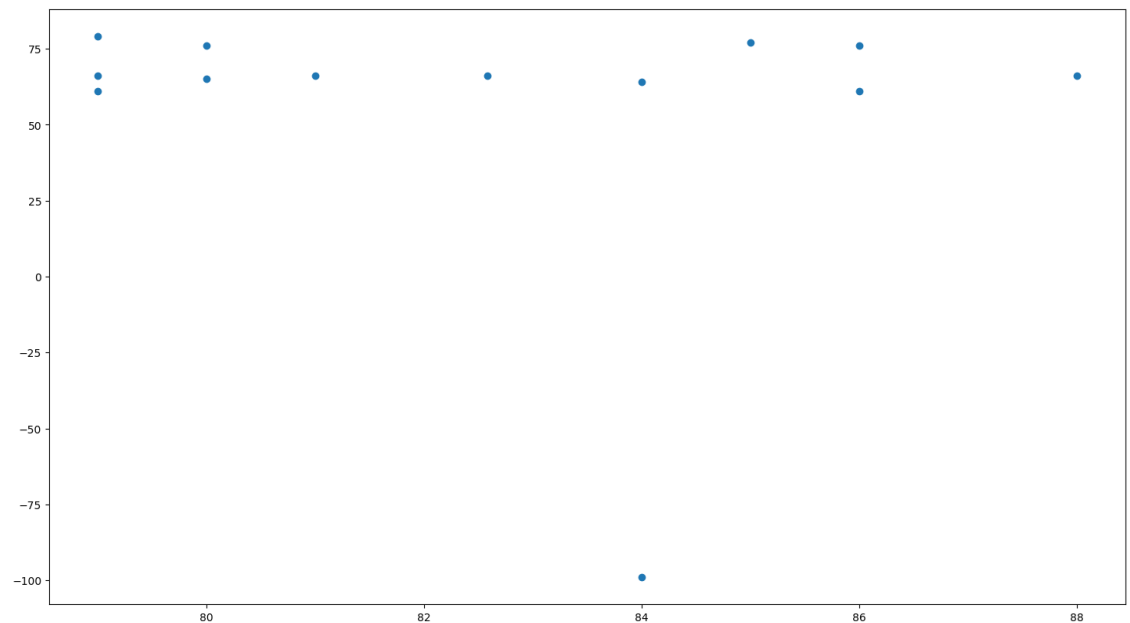
```
In [17]: col=['reading score', 'reading score', 'writing score', 'placement  
df.boxplot(col)
```

Out[17]: <AxesSubplot:>



```
In [18]: print(np.where(df['reading score']>90))  
print(np.where(df['writing score']>90))  
  
(array([], dtype=int64),)  
(array([], dtype=int64),)
```

```
In [19]: fig, ax = plt.subplots(figsize = (18,10))
ax.scatter(df['reading score'], df['writing score'])
plt.show()
ax.set_xlabel('(Proportion non-retail business acres)/(town)')
ax.set_ylabel('(Full-value property-tax rate)/($10,000)')
```



```
Out[19]: Text(4.444444444444452, 0.5, '(Full-value property-tax rate)/($10,000)')
```

```
In [20]: print(np.where((df['reading score'] < 50) & (df['writing score'] > 1)))
print(np.where((df['reading score'] > 85) & (df['writing score'] < 3)))
```

```
(array([], dtype=int64),)
(array([], dtype=int64),)
```

```
In [21]: z = np.abs(stats.zscore(df['reading score']))
```

```
In [22]: print(z)
```

```
0    0.472390
1    0.861418
2    0.527966
3    0.805843
4    1.139295
5    1.139295
6    1.194870
7    0.000000
8    1.806199
9    1.194870
10   0.861418
11   0.472390
12   1.194870
Name: reading score, dtype: float64
```

```
In [23]: threshold = 0.18
```

```
In [24]: sample_outliers = np.where(z < threshold)
```

```
In [25]: sample_outliers
```

```
Out[25]: (array([7], dtype=int64),)
```

```
In [26]: sorted_rscore= sorted(df['reading score'])
```

```
In [27]: sorted_rscore
```

```
Out[27]: [79.0,  
          79.0,  
          79.0,  
          80.0,  
          80.0,  
          81.0,  
          82.58333333333333,  
          84.0,  
          84.0,  
          85.0,  
          86.0,  
          86.0,  
          88.0]
```

```
In [28]: q1 = np.percentile(sorted_rscore, 25)  
         q3 = np.percentile(sorted_rscore, 75)  
         print(q1,q3)
```

```
80.0 85.0
```

```
In [29]: IQR = q3-q1
```

```
In [30]: lwr_bound = q1-(1.5*IQR)  
         upr_bound = q3+(1.5*IQR)  
         print(lwr_bound, upr_bound)
```

```
72.5 92.5
```

```
In [31]: new_df=df
for i in sample_outliers:new_df.drop(i,inplace=True)
new_df
```

```
Out[31]:
```

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.0	64	84.0	2020	2
1	female	71.0	80.0	76	86.0	2018	3
2	female	64.0	81.0	66	81.0	2020	2
3	male	71.0	85.0	77	96.0	2018	1
4	male	68.0	86.0	76	NaN	2021	3
5	female	94.0	86.0	61	100.0	2019	1
6	male	75.0	79.0	66	-99.0	2020	1
8	male	66.0	88.0	66	88.0	2020	3
9	male	70.0	79.0	61	87.0	2021	2
10	female	-99.0	80.0	65	85.0	2021	1
11	male	76.0	84.0	-99	NaN	2020	2
12	female	74.0	79.0	79	98.0	2019	2

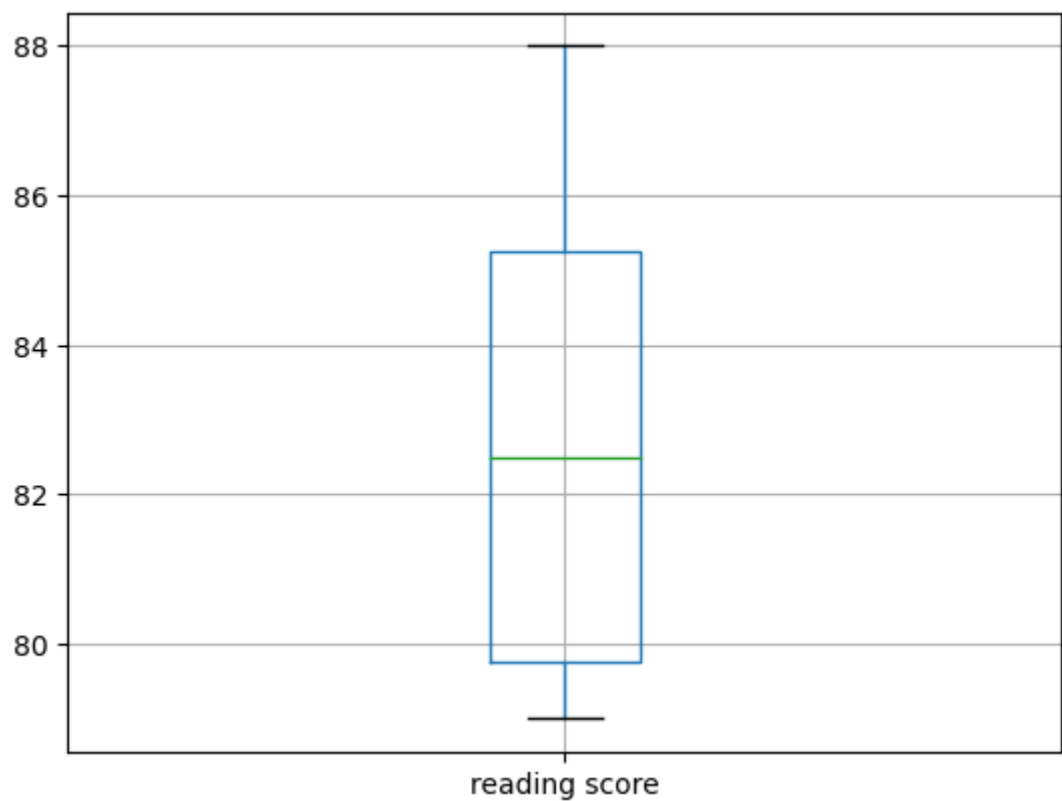
```
In [32]: df=pd.read_csv("D:\StudentPerformance.csv")
```

```
In [33]: df_stud=df
ninetieth_percentile = np.percentile(df_stud['reading score'], 90)
b = np.where(df_stud['reading score']>ninetieth_percentile,
ninetieth_percentile, df_stud['reading score'])
print("New array:",b)
```

```
New array: [84. 80. 81. 85. 86. 86. 79. nan 88. 79. 80. 84. 79.]
```

```
In [34]: col = ['reading score']  
df.boxplot(col)
```

Out[34]: <AxesSubplot:>



```
In [35]: median=np.median(sorted_rscore)  
median
```

Out[35]: 82.58333333333333

```
In [36]: refined_df=df  
refined_df['reading score'] = np.where(refined_df['reading score'] >
```

In [37]:

```
df
```

Out[37]:

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.0	64	84.0	2020	2
1	female	71.0	80.0	76	86.0	2018	3
2	female	64.0	81.0	66	81.0	2020	2
3	male	71.0	85.0	77	96.0	2018	1
4	male	68.0	86.0	76	NaN	2021	3
5	female	94.0	86.0	61	100.0	2019	1
6	male	75.0	79.0	66	-99.0	2020	1
7	female	NaN	NaN	66	95.0	2019	3
8	male	66.0	88.0	66	88.0	2020	3
9	male	70.0	79.0	61	87.0	2021	2
10	female	-99.0	80.0	65	85.0	2021	1
11	male	76.0	84.0	-99	NaN	2020	2
12	female	74.0	79.0	79	98.0	2019	2

In [38]:

```
refined_df['reading score'] = np.where(refined_df['reading score'] <
```

In [39]:

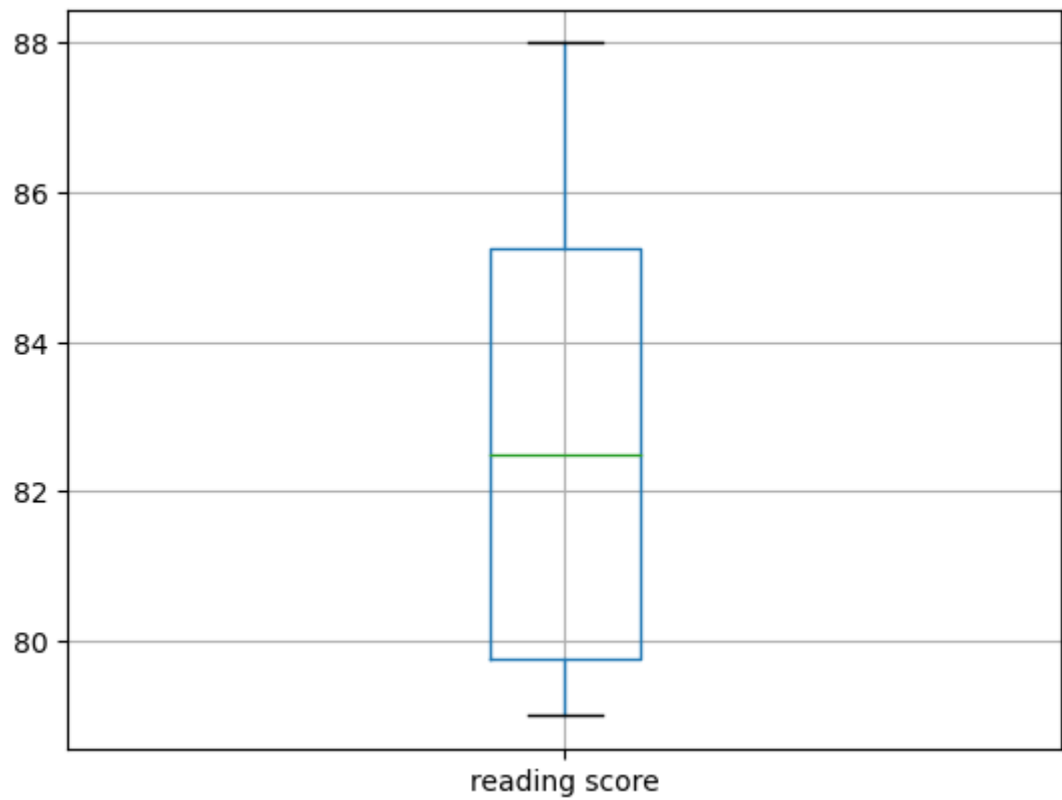
```
df
```

Out[39]:

	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.0	64	84.0	2020	2
1	female	71.0	80.0	76	86.0	2018	3
2	female	64.0	81.0	66	81.0	2020	2
3	male	71.0	85.0	77	96.0	2018	1
4	male	68.0	86.0	76	NaN	2021	3
5	female	94.0	86.0	61	100.0	2019	1
6	male	75.0	79.0	66	-99.0	2020	1
7	female	NaN	NaN	66	95.0	2019	3
8	male	66.0	88.0	66	88.0	2020	3
9	male	70.0	79.0	61	87.0	2021	2
10	female	-99.0	80.0	65	85.0	2021	1
11	male	76.0	84.0	-99	NaN	2020	2
12	female	74.0	79.0	79	98.0	2019	2

```
In [40]: col = ['reading score']
         refined_df.boxplot(col)
```

Out[40]: <AxesSubplot:>

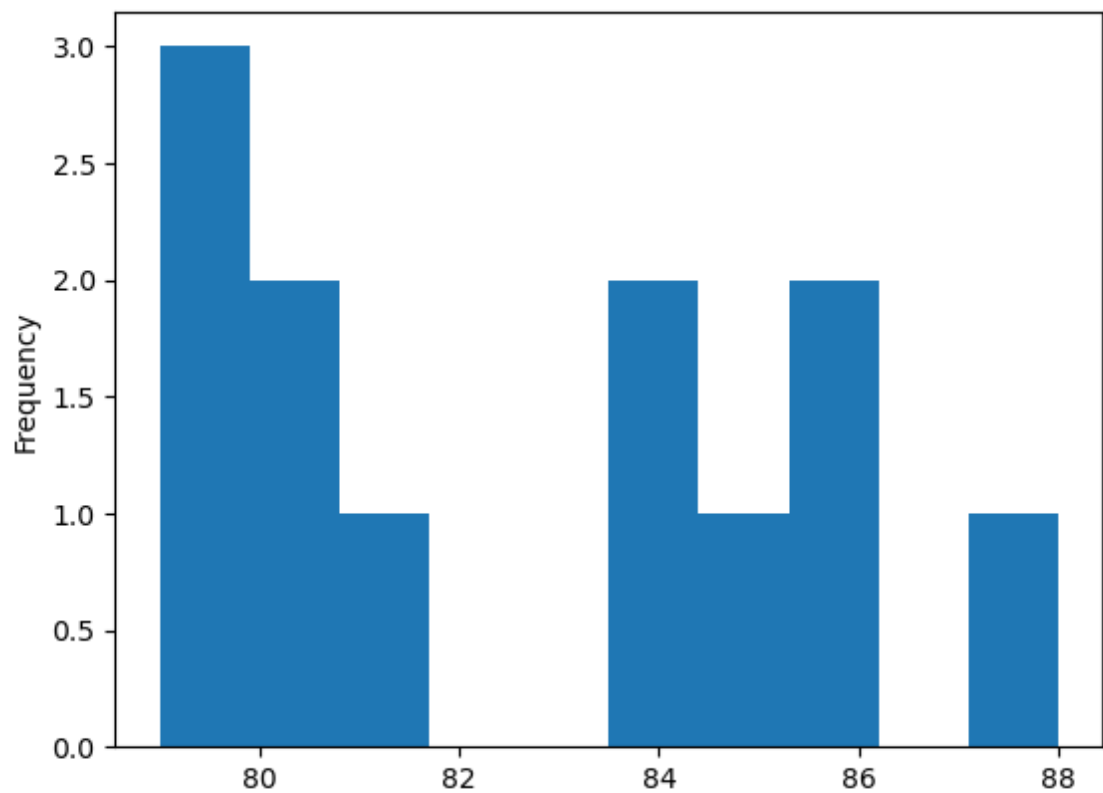


```
In [41]: df
```

Out[41]:

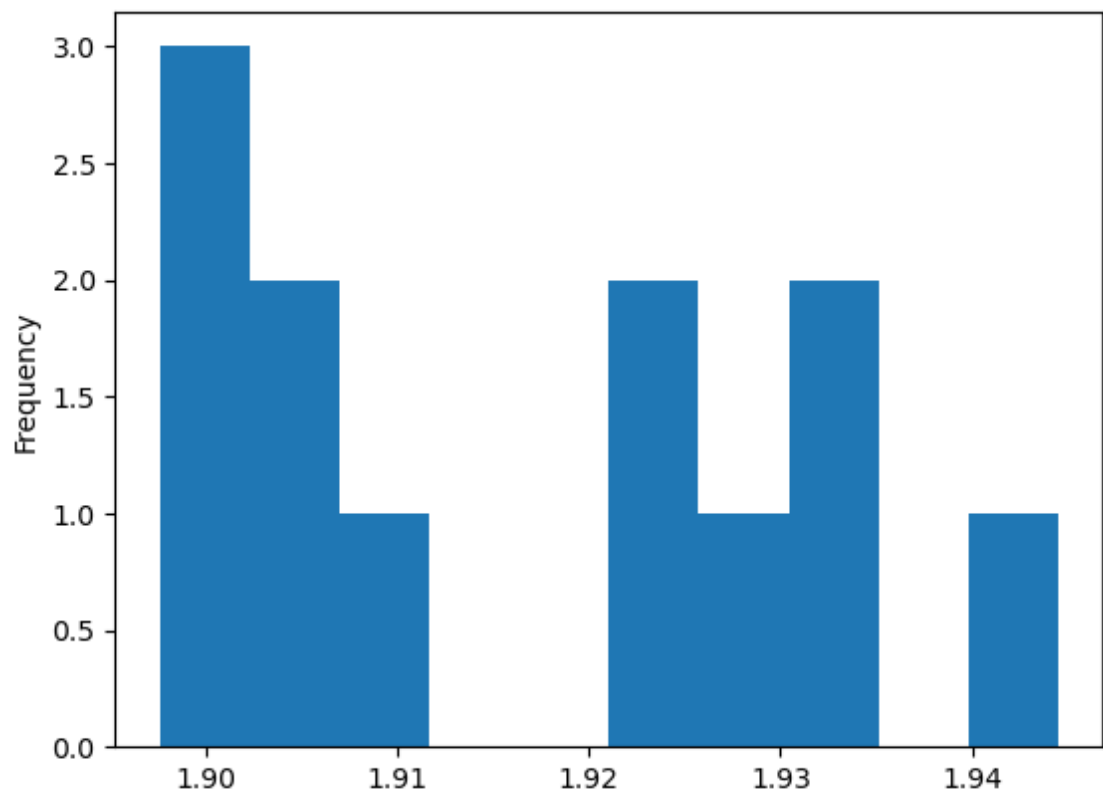
	gender	math score	reading score	writing score	placement score	club join year	placement offer
0	female	63.0	84.0	64	84.0	2020	2
1	female	71.0	80.0	76	86.0	2018	3
2	female	64.0	81.0	66	81.0	2020	2
3	male	71.0	85.0	77	96.0	2018	1
4	male	68.0	86.0	76	NaN	2021	3
5	female	94.0	86.0	61	100.0	2019	1
6	male	75.0	79.0	66	-99.0	2020	1
7	female	NaN	NaN	66	95.0	2019	3
8	male	66.0	88.0	66	88.0	2020	3
9	male	70.0	79.0	61	87.0	2021	2
10	female	-99.0	80.0	65	85.0	2021	1
11	male	76.0	84.0	-99	NaN	2020	2
12	female	74.0	79.0	79	98.0	2019	2

```
In [42]: import matplotlib.pyplot as plt
new_df['reading score'].plot(kind='hist')
df['log_math'] = np.log10(df['reading score'])
```



```
In [43]: df['log_math'].plot(kind='hist')
```

Out[43]: <AxesSubplot:ylabel='Frequency'>



In []:

In []: